DSC 140A - Midterm 01 Review

Problem 1.

a) Suppose $f(\vec{x})$ is a **convex** function of $\vec{x}$ and that the gradient of f at the point $\vec{x}^{(1)} = (0,5)^T$ is $(-3,5)^T$.

Let $\vec{x}^* = (x_1^*, x_2^*)$ be the minimizer of f; you can assume that $\vec{x}^* \neq \vec{x}^{(1)}$, which implies that $f(\vec{x}^*) < f(\vec{x}^{(1)})$. True or False: it must be the case that $x_2^* \geq 5$.

b) Explain in your own words why convexity is such a desirable property for a loss function when using gradient descent. What can go wrong when the function is *not* convex? Your answer should address the concept of local vs. global minima.

Problem 2.

Let $f_1(x)$ be a convex function and let $f_2(x)$ be a non-convex function. Define the new function $f(x) = f_1(x) + f_2(x)$. True or False: $f(x)$ must be non-convex.

○ True

○ False

Problem 3.

Recall that a subgradient of the absolute loss is:

$$\begin{cases} \operatorname{Aug}(\vec{x}), & \text{if } \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y > 0, \\ -\operatorname{Aug}(\vec{x}), & \text{if} \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y < 0, \\ \vec{0}, & \text{otherwise.} \end{cases}$$

Suppose you are running subgradient descent to minimize the risk with respect to the absolute loss in order to train a function $H(x) = w_0 + w_1 x_1 + w_2 x_2$ on the following data set:

x_1	x_2	y
2	5	6
1	2	3

Suppose that the initial weight vector is $\vec{w} = (0,0,0)^T$ and that the learning rate $\eta = 1$. What will be the weight vector after one iteration of subgradient descent?

Problem 4.

Let $X = \{\vec{x}^{(i)}, y_i\}$ be a data set of n points where each $\vec{x}^{(i)} \in \mathbb{R}^d$.

Let $Z = \{\vec{z}^{(i)}, y_i\}$ be the data set obtained from the original by standardizing each feature. That is, if a matrix were created with the i-th row being $\vec{z}^{(i)}$, then the mean of each column would be 0, and the variance would be 1.

a) Suppose linear predictors H_1 and H_2 are fit on X and Z by minimizing the MSE, respectively.

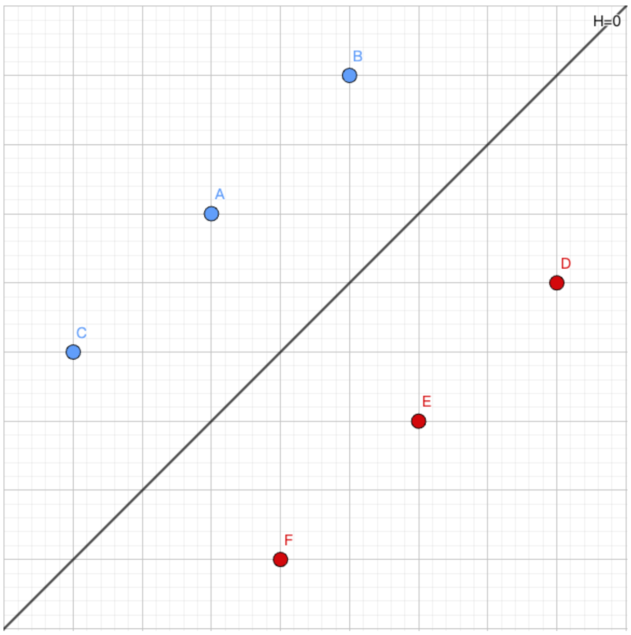
True or False: $H_1(\vec{x}^{(i)}) = H_2(\vec{z}^{(i)})$ for every $i = 1, \ldots, n$.

b) Suppose that X and Z are both linearly-separable. Suppose Hard-SVMs H_1 and H_2 are trained on X and Z, respectively.

True or False: $H_1(\vec{x}^{(i)}) = H_2(\vec{z}^{(i)})$ for every $i = 1, \ldots, n$.

Problem 5.

Consider the image below:



The blue points have label $+1$, and the red points have label -1. Suppose H is a linear prediction function, and when H is applied to the point A in the above image, $H(\vec{x}) = 4$. The black line in the middle of the image is the decision boundary $H = 0$.

You may assume that H is exactly in the middle of the points, and that all points are equidistant from the decision boundary.

a) What is the mean square loss of this prediction function, H?

b) True or false: there exists a linear prediction function H which has a mean square loss of zero on this data.

Problem 6.

Suppose a line of the form $H(x) = w_0 + w_1 x$ is fit to a data set of points $\{(x_i, y_i)\}$ in $\mathbb{R}^2$ by minimizing the mean squared error. Let the mean squared error of this predictor with respect to this data set be E_1.

Next, create a new data set by adding a single new point to the original data set with the property that the new point lies *exactly* on the line $H(x) = w_0 + w_1 x$ that was fit above. Let the mean squared error of H on this new data set be E_2.

Which of the following is true?

- ○ $E_1 < E_2$
- ○ $E_1 = E_2$
- ○ $E_1 > E_2$

Problem 7.

Consider a binary classification problem in $\mathbb{R}^2$ with labels $y \in \{-1, 1\}$. Suppose we train a linear classifier with weight vector $\vec{w} = (w_0, w_1, w_2)$ and use the predictor $\operatorname{sign}(\operatorname{Aug}(\vec{x}) \cdot \vec{w})$.

a) Explain in 2–3 sentences what the *decision boundary* is geometrically, and what role the *magnitude* of $H(\vec{x}) = \operatorname{Aug}(\vec{x}) \cdot \vec{w}$ plays in a linear classifier's predictions. In particular, what does it mean for a correctly-classified point to have a large $|H(\vec{x})|$ vs. a small $|H(\vec{x})|$?

b) Imagine a different a $H(x)$ that uses nonlinear features, for example $H(x) = \vec{w}^T \phi(x)$ where $\phi(x) = [x, x^2]^T$. Would our decision boundary still be a plane in our data space? Explain.

c) Suppose $\vec{w} = (w_0, w_1, w_2) = (1, 2, -1)$ (where $w_0 = 1$ is the bias). Consider the following four points and their true labels:

Point $\vec{x}^{(i)}$	True label y_i
$(1,\ 3)$	$+1$
$(2,\ 1)$	-1
$(-1,\ 2)$	$+1$
$(0,\ -1)$	-1

For each point, compute $H(\vec{x}^{(i)}) = \operatorname{Aug}(\vec{x}^{(i)}) \cdot \vec{w}$, state whether it is correctly classified, and compute the **perceptron loss** ℓ_{tron} and the **0-1 loss** $\ell_{\text{0-1}}$ for each point.

d) The 0-1 loss directly measures what we care about (misclassification), yet we use other loss functions (like perceptron loss or hinge loss) during training. Why don't we use 0-1 loss?

Problem 8.

In class, we discussed how L1 regularization encourages sparse solutions and can be seen as a method for feature selection. In this problem, we will explore why L1 regularization promotes sparsity from the perspective of gradient descent.

a) First, write down the partial derivatives of the L1 and L2 regularization terms with respect to a specific weight w_j (you may ignore the case where $w_j = 0$, as the gradient might be undefined there).

- The L1 regularization term is given by:

$$R_1(\vec{w}) = \lambda \sum_{j=1}^{d} |w_j|$$

- The L2 regularization term is given by:

$$R_2(\vec{w}) = \lambda \sum_{j=1}^{d} w_j^2$$

b) Based on these derivatives, which regularizer is more effective at pushing w_j to zero?

Hint: Consider the behavior of the gradients when w_j is already small. For simplicity, assume that the partial derivative $\frac{\partial}{\partial w_j}$ of the Mean Squared Error (MSE) term is zero.

Problem 9.

Consider the following two scenarios:

- **Scenario A:** You apply the feature map $\vec{\phi}(x_1, x_2) = (x_1,\ x_2,\ x_1^2,\ x_2^2,\ x_1 x_2)^T$ to your data and train a least squares classifier in feature space.
- **Scenario B:** You skip the feature map entirely and train a least squares classifier directly on $(x_1, x_2)^T$.

A classmate says: *"Scenario A is strictly better than Scenario B — more features always means a better model."* Another classmate disagrees: *"Scenario A is not always better. There are cases where adding these extra features could actually hurt you."*

Which classmate is correct? Describe a concrete situation where Scenario A would perform **worse** than Scenario B on new, unseen data. In your answer, explain the mechanism by which this happens.

Problem 10. (2 points)

Consider the data set: $\vec{x}^{(1)} = (1, 0, 2)^T$ $\vec{x}^{(2)} = (-1, 0, -1)^T$ $\vec{x}^{(3)} = (1, 2, 1)^T$ $\vec{x}^{(4)} = (1, 1, 0)^T$

Suppose a prediction function $H(\vec{x})$ is learned using kernel ridge regression on the above data set using the kernel $\kappa(\vec{x}, \vec{x}') = (1 + \vec{x} \cdot \vec{x}')^2$ and regularization parameter $\lambda = 3$. Suppose that $\vec{\alpha} = (-1, -2, 0, 2)^T$ is the solution of the dual problem.

a) What is the (2,3) entry of the kernel matrix?

1

b) Let $\vec{x} = (1, 1, 0)^T$ be a new point. What is $H(\vec{x})$?

Problem 11.

Let $\{\vec{x}^{(i)}, y_i\}$ be a data set of n points, with each $\vec{x}^{(i)} \in \mathbb{R}^d$. Recall that the solution to the kernel ridge regression problem is $\vec{\alpha} = (K + n\lambda I)^{-1}\vec{y}$, where K is the kernel matrix, I is the identity matrix, $\lambda > 0$ is a regularization parameter, and $\vec{y} = (y_1, \ldots, y_n)^T$.

Suppose kernel ridge regression is performed with a kernel κ that is a kernel for a feature map $\vec{\phi} : \mathbb{R}^d \to \mathbb{R}^k$.

What is the size of the kernel matrix, K?

- ◯ $d \times d$
- ◯ $k \times k$
- ◯ $n \times n$
- ◯ $n \times d$

Problem 12.

In lecture, we saw that SVMs minimize the (regularized) risk with respect to the *hinge* loss. Now recall, the 0-1 loss, which was defined in lecture as:

$$L(\vec{x}^{(i)}, y_i, \vec{w}) = \begin{cases} 0, & \text{if } y_i = \text{sign}(\vec{w} \cdot \text{Aug}(\vec{x}^{(i)})), \\ 1, & \text{otherwise.} \end{cases}$$

a) Suppose a **Hard** SVM is trained on linearly-separable data. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

- ◯ True

◯ False

b) Suppose a **Soft** SVM is trained on linearly-separable data using an unknown slack parameter C. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

◯ True

◯ False

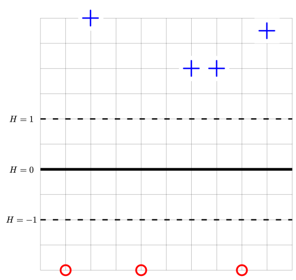
c) Suppose a **Soft** SVM is trained on data that is **not** linearly separable using an unknown slack parameter C. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x})$.

◯ True

◯ False

Problem 13.

The image below shows a linear prediction function H along with a data set; the "$\times$" points have label $+1$ while the "$\circ$" points have label -1. Also shown are the places where the output of H is 0, 1, and -1.



True or False: H could have been learned by training a Hard-SVM on this data set.

◯ True

◯ False

Problem 14.

Consider the data set shown below:

x	y
-1	2
-3	3
2	4
4	7
5	7
6	10

What is the predicted value of y at $x = 3$ if the 3-nearest neighbor rule is used?